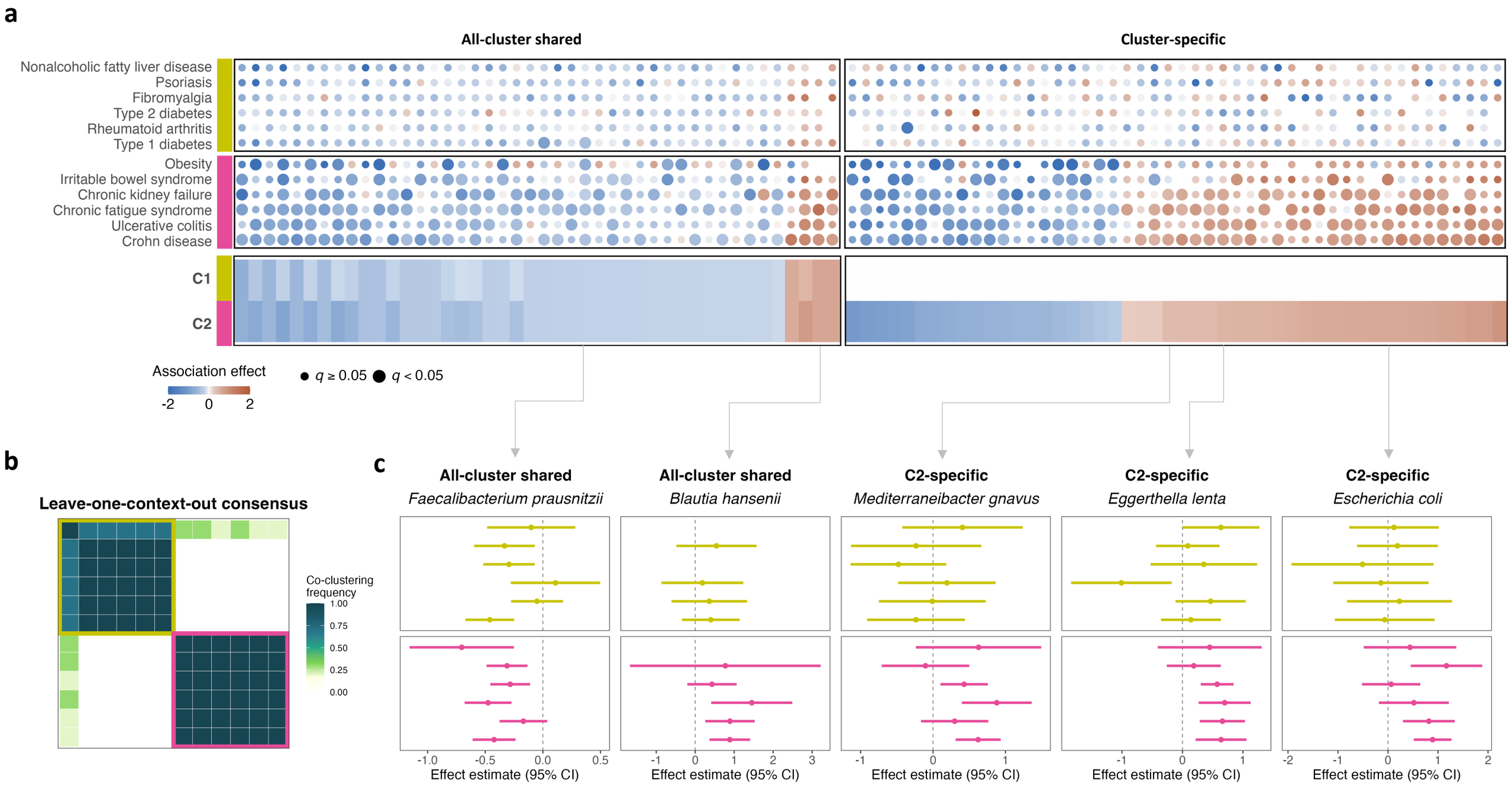

a
All-cluster shared
Cluster-specific
Nonalcoholic fatty liver disease
Psoriasis
Fibromyalgia
Type 2 diabetes
Rheumatoid arthritis
Type 1 diabetes
Obesity
Irritable bowel syndrome
Chronic kidney failure
Chronic fatigue syndrome
Ulcerative colitis
Crohn disease
C1
C2
Association effect
-2
0
2
q ≥ 0.05
q < 0.05
b
Leave-one-context-out consensus
Co-clustering frequency
1.00
0.75
0.50
0.25
0.00
c
All-cluster shared
Faecalibacterium prausnitzii
All-cluster shared
Blautia hansenii
C2-specific
Mediterraneibacter gnavus
C2-specific
Eggerthella lenta
C2-specific
Escherichia coli
Effect estimate (95% CI)
d
Comparison of clustering assignments
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3
SKM
SHC
e
Number of selected signatures
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3
SKM+FE
SHC+FE
Melody
44
48
39
59
28
117
2
33
16
107
16
107
95
0
20
40
60
80
100
120
Jaccard overlap (vs SMESH-PALM)
1.00
1.00
0.11
0.16
0.18
0.24
0.12
0.09
0.27
0.09
0.27
0.09
0.00
0.25
0.50
0.75
1.00
All-cluster shared
Context-dependent